

MDHNT Level-1 Benchmarks

Version 1 — five questions, pre-registered grading

August 2026

These questions accompany *Multi-Dimensional Harmonic Number Theory, for Dummies*. They are answerable with that document, general mathematics, and standard software. Some have known answers (they grade the *transmission* of the toolkit); some have answers nobody yet knows (they grade the *toolkit*). Each states its own success and failure conditions in advance. Report format for all: verdict with numbers, your derivations, your code, and a build log of every definition you had to supply yourself.

Q1 — The invariant (calibration; answer known)

Build the four-line ledger instrument of the guide’s clock-chart recipe. Derive the general k -event cell residue, and from it an *occupancy-free invariant*: a quantity built from the residue and the event phases that is confined to a lattice in every cell, regardless of occupancy or history. Verify it numerically over at least 10^3 cells. *Grading*: the invariant is exact; any deviation beyond quadrature error in any cell is a fail — of your build, or of the guide, and your report should say which.

Q2 — The pair question (measurement; answer unknown, and the question is which *mechanism* survives)

For cells containing exactly two events: recover the unordered phase pair from ledger moments alone (no zero locations), and measure the distributions of the separation $D = |\tau_1 - \tau_2|$ and the barycenter, over every two-event cell available to you. Then decide between three pre-registered hypotheses — and note carefully that the second is *not* established mathematics:

- (a) *Independence*: phases i.i.d. uniform. Triangular D -density, $\rho(0) = 2$, $\mathbb{E}[D^2] = 1/6$.
- (b) *Interaction* (“*repulsion*”): sine-kernel-like local statistics, $\rho(D) \propto D^2$ as $D \rightarrow 0$. This is the random-matrix conjecture; it is supported numerically in the literature and *proven nowhere* beyond a band-limited regime.
- (c) *Registration displacement*: a two-event cell is a bookkeeping event — one native occupant plus one member displaced across a boundary from an adjacent cell — so close pairs are rare and gaps large *without any interaction at all*. This hypothesis makes predictions (b) does not: two-event cells should be adjacent to event-free cells with the correlation forced by the ledger (the census identity $\#empty = \#double + 2 \cdot \#triple$ is exact by

telescoping — verify it), and, conditional on such adjacency, the pair member nearer the shared boundary is the displaced one, so its phase should hug that boundary.

Grading: the small- D exponent with error bars, the barycenter distribution reported as found, and above all the *adjacency statistic*: it is the discriminator, since (b) and (c) fit the same gap histogram but only (c) ties pair geometry to neighboring cells' occupancy. A verdict for (c) would demote “repulsion,” for this ledger, from phenomenon to chart-name for registration — which is why the question is worth asking.

Q3 — The sharp null (design + measurement; answer unknown)

The guide pre-registers a two-tier calibration and admits nobody has run the sharp tier: a random *completely multiplicative* model (Steinhaus: independent uniform phases on the primes, extended multiplicatively), which keeps the multiplicative skeleton and destroys only the arithmetic. Design a defensible ledger instrument for this model — the design is graded, and the central difficulty (what plays the role of the event set for a model with no functional equation) is yours to resolve and defend — and measure B_N . *Grading:* either verdict is publishable. Bounded B_N means the boundedness claim is about multiplicativity, not arithmetic, and the guide's calibration section gets rewritten. Growing B_N sharpens the claim that registration is arithmetic. An honest “the question is ill-posed as stated, and here is why” with a repaired statement also passes.

Q4 — Dimension parity (pencil and paper; answer derivable)

The guide's carry-counting example computes the archimedean exponent $A(g)$ of the tensor tower of $H^1(E)$, E an elliptic curve, from its Hodge diamond. Do the same, directly, for an abelian surface A ($h^{1,0} = h^{0,1} = 2$) at $g = 3$: compute the exponent from the diamond of $H^1(A)^{\otimes 3}$, compare with the elliptic value, and state — with proof — the sign conclusion for the tensor tower of an abelian variety of *any* dimension d at every odd grade. *Grading:* the d -scaling law and the parity dichotomy it forces; a normalization subtlety that breaks the naive scaling, if you find one, is worth more than the confirmation.

Q5 — The mechanism (structural; answer unknown)

The zeta ledger's running cell-residue sum B_N is empirically flat across thousands of cells. Locate the cancellation: quantify how much of the per-cell residue variance is carried by within-cell phase configurations versus by the sequence of integer boundary values s_m , and identify a measurable property of the s_m sequence sufficient to account for the flatness. Propose and test at least one statistic of your own design. *Grading:* a decomposition with numbers, a named sufficient property, and a pre-registered test of it. “The variance lives here, and here is the correlation structure that cancels it” is a pass; a mechanism that would also make the null flat is an automatic fail (it proves too much).

Protocol note. Attempt any one question cold, from the guide and this sheet alone. Your build log — what the documents failed to give you — is data of equal value to your verdict, and will be treated as such.